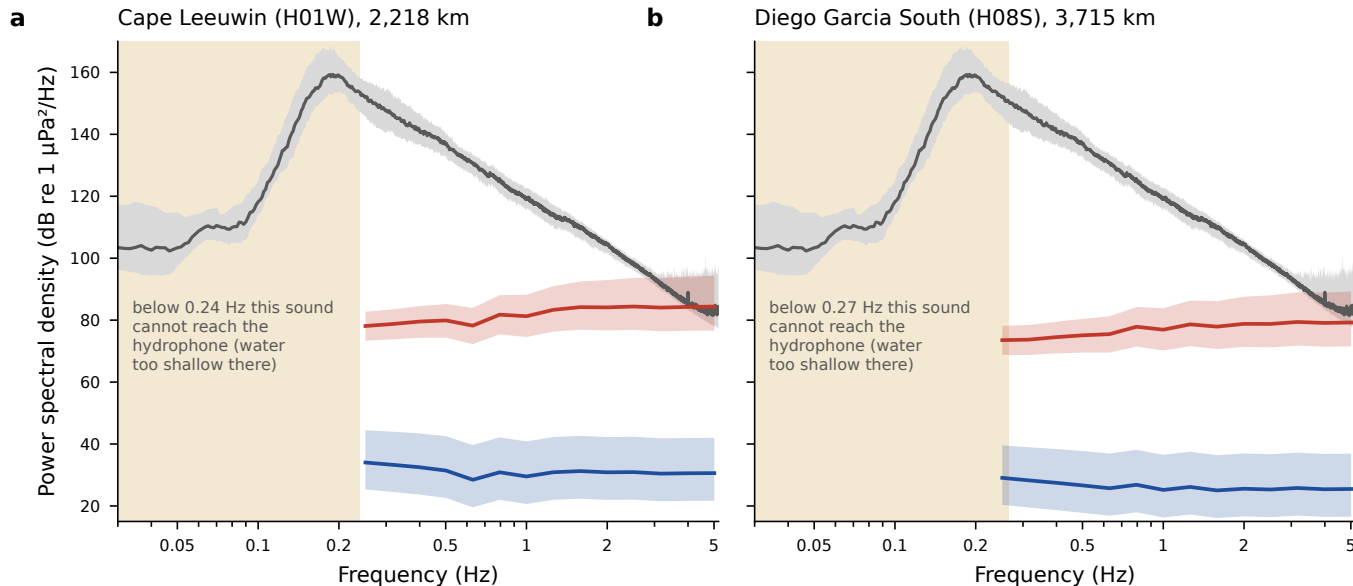


Low-frequency sound from the impact is far below ocean noise below 2 Hz at both stations

■ ocean pressure noise, 10-90 %
— ocean pressure noise, median
— impact signal: steep fast entry (F-35A type, 175 t)
— impact signal: controlled ditching (175 t)



What the chart shows: the predicted low-frequency sound (0.03-5 Hz, the band of the 'acoustic-gravity' waves, which fill the full water depth) from two impact types, against ocean pressure noise measured on the deep sea floor of the south-west Indian Ocean in March 2013. Below about 0.25 Hz the sound cannot reach either hydrophone, because the water at the hydrophone is only about 1.5 km deep. From 0.25 to 2 Hz the noise is 20-110 dB above the signal. Only the steep fast entry comes near the noise, at 2.5-5 Hz, which is the low end of the normal sound band that our main model already covers. A ditching is 40-110 dB below the noise at all frequencies shown. This is an estimate to the nearest order of magnitude, not a calibrated prediction.

Technical (exploratory, pre-registered agw_scenarios.py): vertical surface-force impulse $J = m v_z$, single-pole roll-off τ ; (a) J 2.4-6.3e7 N s, τ 0.1-0.3 s; (b) J 0.42-1.45e6 N s, τ 1-3 s; line = geometric-mean J , mean τ . Isovelocity rigid-bottom normal modes (Jensen et al. 2011, ch. 5), adiabatic, GEMCO_2026 depths; modes cut off on the path are lost (no tunnelling); arrival spread ≥ 10 s; model ± 10 dB, not shown. Impact 37.23° S 89.58° E. Noise: RHUM-RUM differential pressure gauges RR50/RR40/RR38/RR34 (RESIF YV, 4.1-4.8 km), 1-14 Mar 2013 00-03 UTC, 56 blocks, peak 0.19 Hz; sea-floor, not hydrophone depth; gain ± 3 dB. IMS response (-47 dB at 0.1 Hz) acts on signal and noise alike; instrument self-noise not checked. Horizontal momentum not modelled.